

C330

METAL PLATE EMBOSSER



OPERATOR MANUAL

Revision 1.0



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1. INTRODUCTION

1.1. Packing list

The machine is delivered in a cardboard box.

 **NOTE: move the machine always in two persons.**

Machine dimension: 61 x 43 x 25 cm

Machine weight: 29 Kg

1. Open the box and check the packing list:

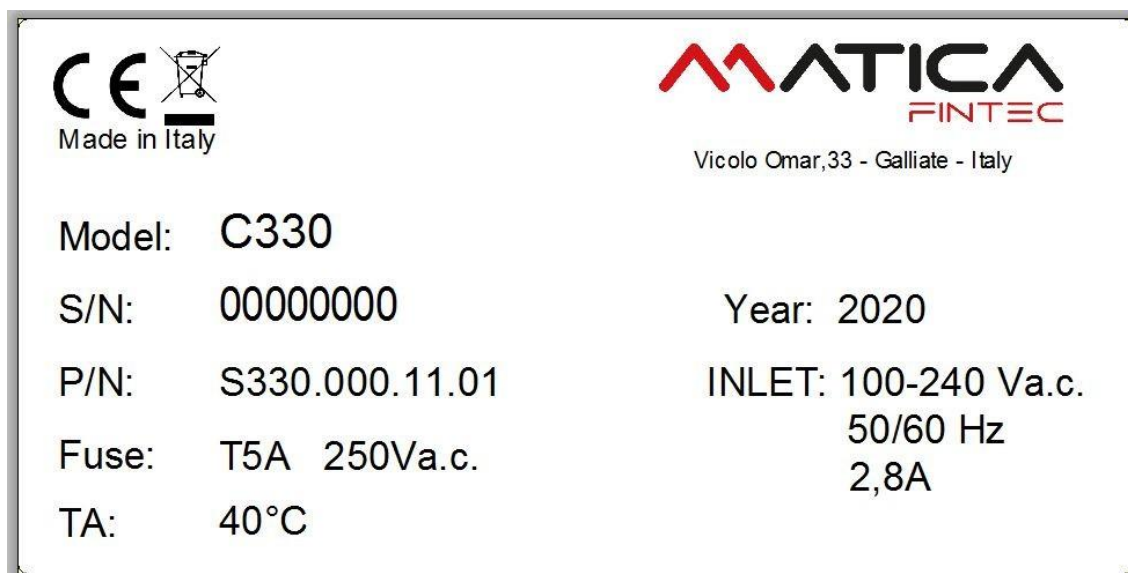
- Power cord
- USB cable
- USB Pen Drive

2. Remove the top cushion

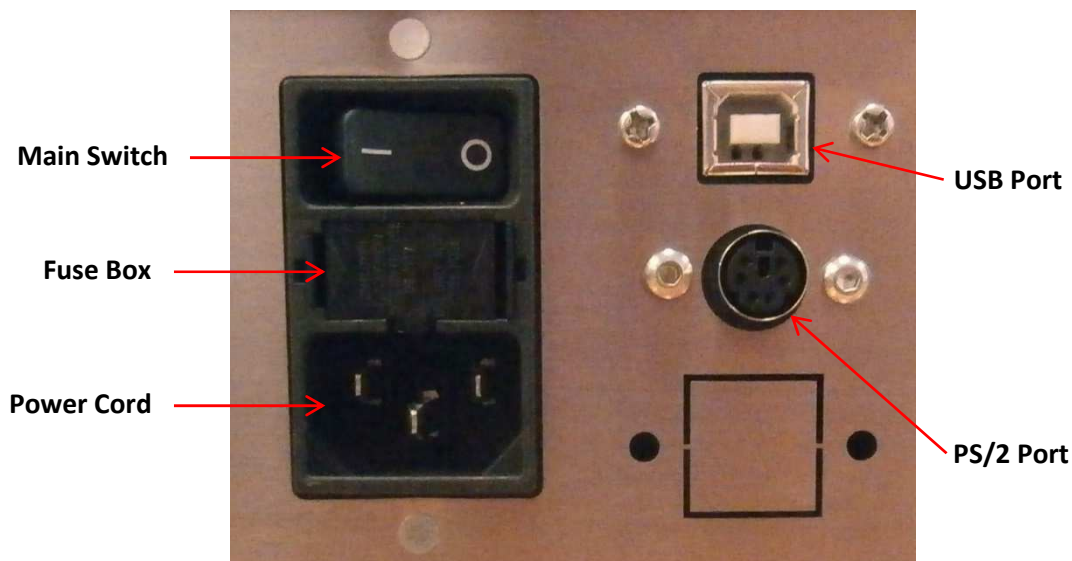
3. Put the machine on a table

1.2. Installation

- Verify that the voltage marked on the machine label corresponds with your country voltage supply.



- Connect the power cord.
- Connect the USB cable
- Power on the machine.



1.3. Console

The C330 console is made by:

1. LCD display (2 lines per 40 characters)
2. Key Clear used to clear the error condition



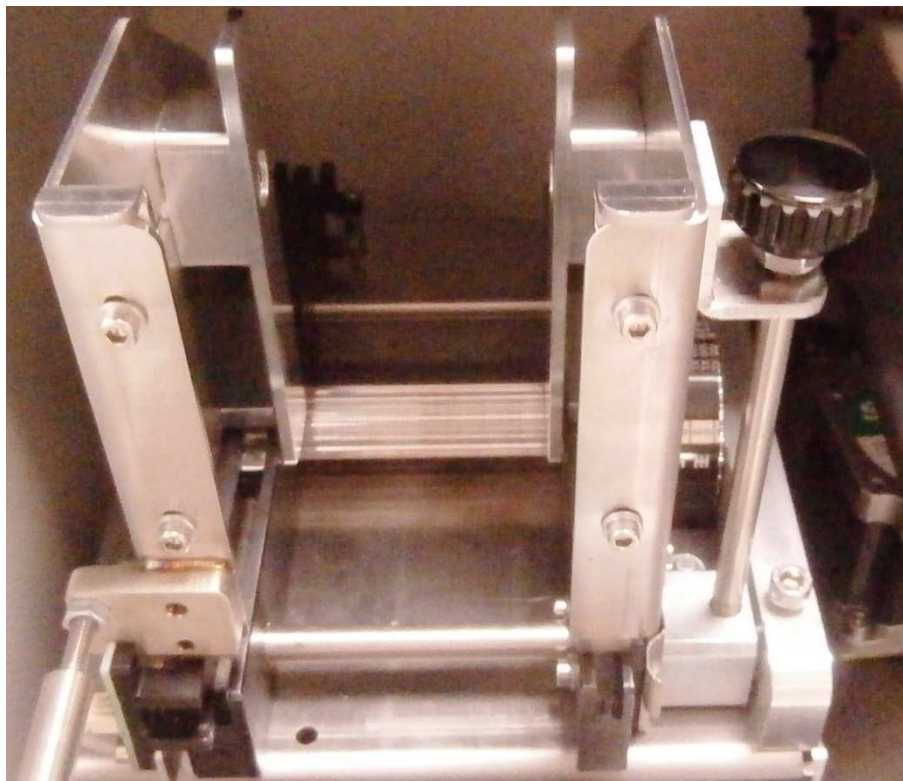
1.4. Configuration

C330 is provided with the following features:

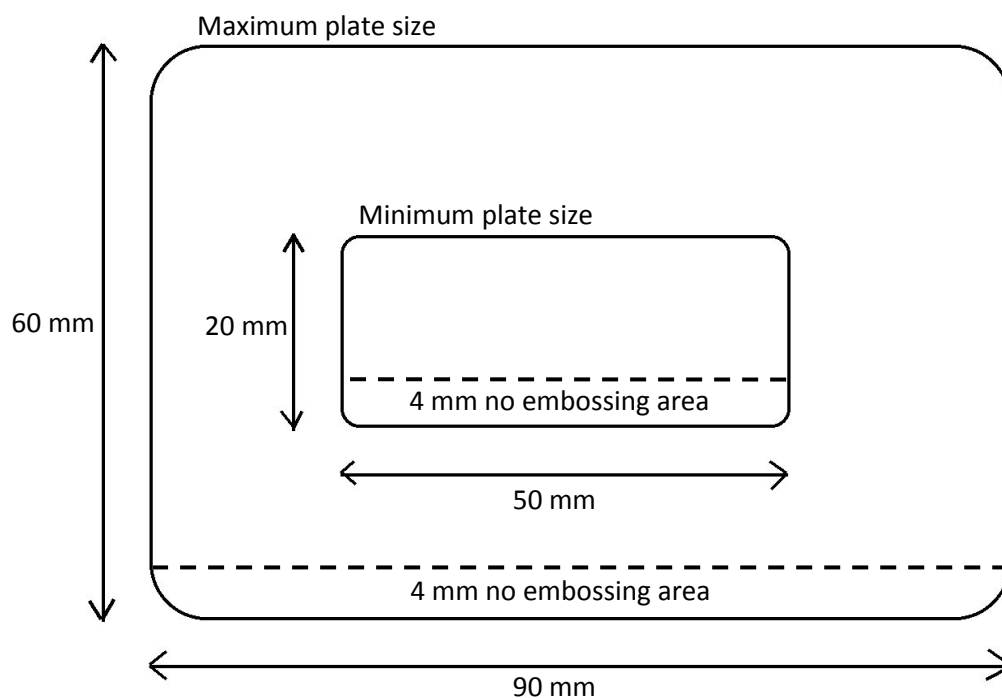
- Adjustable feeder
- Embosser
- Unloader

2. ADJUSTABLE FEEDER

C330 is provided with an adjustable feeder.

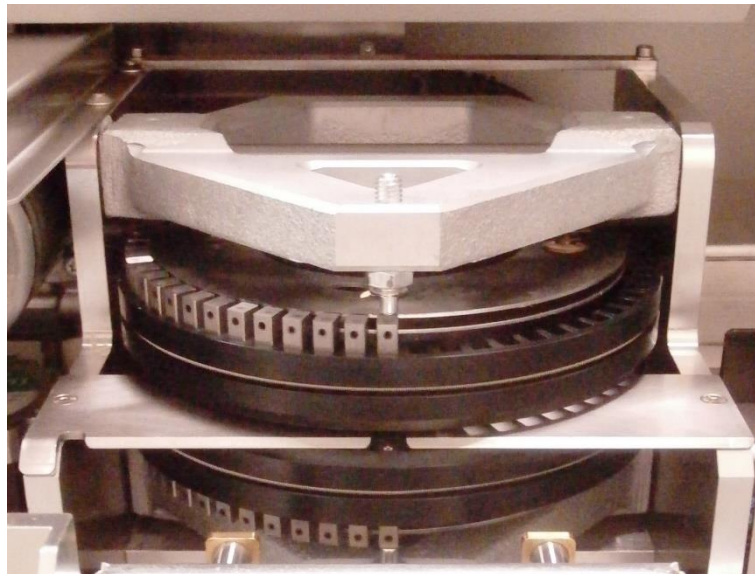


The feeder height is 95 mm, it has the capacity of 190 plates (0,5 mm thickness) and it can manage different thickness from 0,3 mm to 0,8 mm. The plate dimension can be changed unscrewing the black knob for X-axis or moving the two plate guides for Y-axis. The minimum dimension for a plate is 50 x 20 mm and the maximum one is 90 x 60 mm.



3. EMBOSSE

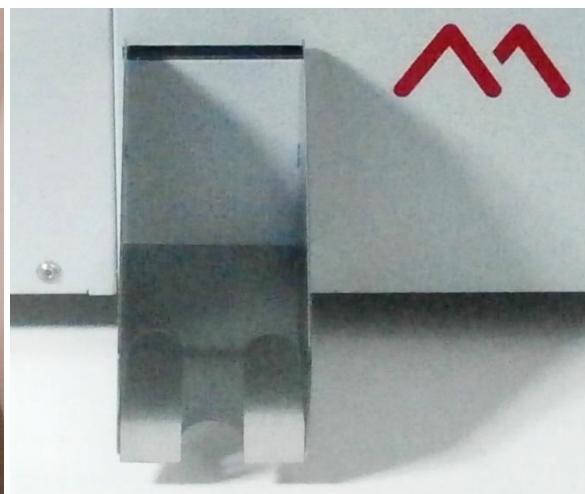
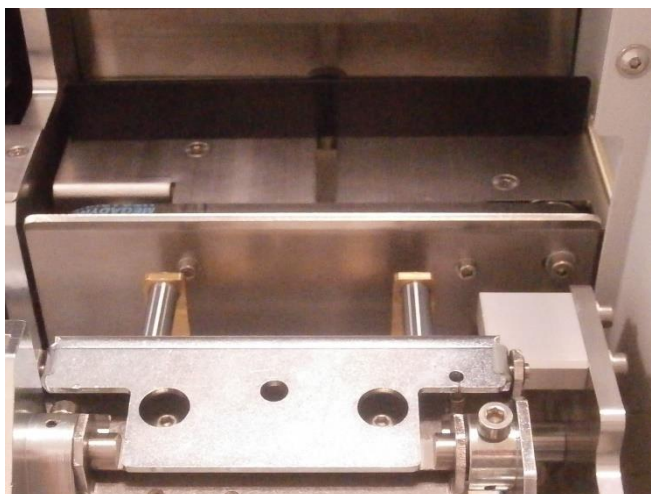
C330 is provided with fixed embosser module as standard configuration and uses embossing drum with 60 characters.



The embossing area is 80 x 56 mm, leaving 4 mm from the bottom of the plate for the clamp grip.

4. UNLOADER

C330 is provided with a side eject unloading module, the plate is stacked in the output box.



5. Power-on the C330

Before power on the machine check the top cover is closed.

If cover is open the machine do not start, and if the cover is open during the working modality all the movements will be immediately STOP and a COVER OPEN message will be show on display.

Machine operator ONLY with cover closed.

Powering on the machine, the LCD display will show:

POWER-ON	C330 Ver. C-x-xxx
PRESS CLEAR or ESC to CONTINUE	

Press CLEAR button to start the INITIALIZATION cycle that will homing (restore) the Punch, X, Y and Drum mechanism; and execute and eject cycle to eject the plate in the clamp and to open the clamp to then be ready to emboss plates.

During the initialization cycle the display show the Protocol and the baud rate configured

INITIALIZATION	Prot:Xon-Xoff Baud:57600
----------------	--------------------------

At the end will be show the machine name and the READY status, and the total counter of plates produced.

C330 METAL EMBOSSEY BY MATICA FINTEC -ITALY
READY Counter:00000100

5.1. Working modality

During working the display will show FEEDING, EMBOSSING and EJECTING to indicate the current cycle in process.

5.2. Error condition

In case of error the display will show the error number and the message:

Exx	FEEDER EMPTY
PRESS CLEAR or ESC to CONTINUE	

- Open the cover and remove the cause of the error
- Close the cover
- Press CLEAR button to continue the process
- In case of feeding error, the machine will keep retry until the plate will be correctly loaded and embossing will start.
- In case of error during the embossing cycle the plate will be eject and NOT DUPLICATED.

6. Connect to PC and install the Generic Printer Driver

6.1. Install COM PORT driver

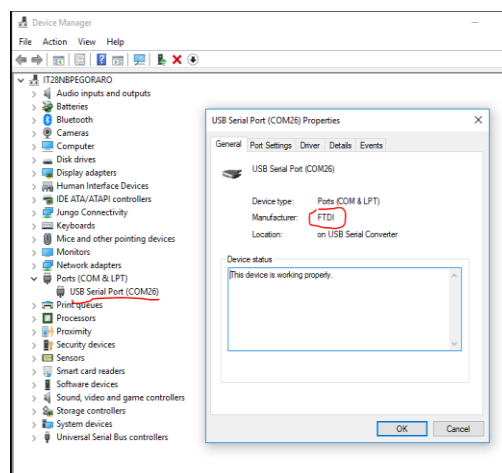
Connecting the USB cable to PC and power on the machine. Windows automatically detect the new COM and install the proper FTDI driver.

Use Device Manager to check the COMxx number assigned.

- Be sure the COMxx is installed
- If the COM is not installed, then install using the file C:\MATICA\...\Installation Files\USB2COM FTDI VCP Diver - CDM21228_Setup
- Remember the COMxx number we will need to install the Generic printer driver.

On your PC Windows will sign and COMxx number that will be keep the same changing USB plugs on your PC. So, you don't need to remember in which USB you plug the first time the machine.

If you connect another machine (different SN) Windows will assign a new COMxx number. Via Advance PORT configuration is possible to change the COMxx number



6.2. Install Generic printer driver (Windows 10)

The C330 can be easily managed via printer driver, like a standard printer, sending text to emboss using a very intuitive editing like Notepad or Word.

1. Go to Printers & scanners
2. Add a printer
wait windows to search for new printers
when done go to the end of the list and
3. Select "The printer that I want isn't listed"
4. Add Printer
5. Select "Add a local printer"

Printers & scanners

Add printers & scanners



Add a printer or scanner



Add Printer

Find a printer by other options

- ☐ My printer is a little older. Help me find it.
- ☐ Find a printer in the directory, based on location or feature
- ☐ Select a shared printer by name

Example: \\computename\printername or
http://computename/printers/printername/.printer

- ☐ Add a printer using a TCP/IP address or hostname
- ☐ Add a Bluetooth, wireless or network discoverable printer
- ☒ Add a local printer or network printer with manual settings

6. Choose a printer port
7. Select the COMxx of the machine

Choose a printer port

A printer port is a type of connection that allows your computer to exchange information

☒ Use an existing port:

☐ Create a new port:

Type of port:

- LPT1: (Printer Port)
- LPT1: (Printer Port)
- LPT2: (Printer Port)
- LPT3: (Printer Port)
- COM1: (Serial Port)
- COM2: (Serial Port)
- COM3: (Serial Port)
- COM5: (Serial Port)
- COM6: (Serial Port)
- COM7: (Serial Port)

8. Install the printer driver

Install the printer driver



Choose your printer from the list. Click Windows Update to see more models.

To install the driver from an installation CD, click Have Disk.

9. Select

- Manufacturer → **Generic**
- Printers → **Generic / Text Only**

Manufacturer: Generic

Printers: Generic / Text Only

Generic IBM Graphics 9pin

Generic IBM Graphics 9pin wide

MS Publisher Color Printer

This driver is digitally signed.

[Tell me why driver signing is important](#)

Windows Update Have Disk...

10. Give a name at the printer

Type a printer name

Printer name: C330

This printer will be installed with the Generic / Text Only driver.

11. Configure the COM port in the printer driver

12. Go to C330 Properties (or the name you define)

13. Go to COMxx Properties and select:

- Bits per second → **57600**
- Flow control → **Xon / Xoff**

We are now ready to produce plates.

note: Printer driver is normally hold the COMxx in exclusive, blocking other application (like diagnostic tool) to access the machine. To FREE the COMxx simply assing the printer (C330) to FILE.

C330 Properties

Security General Device Settings Ports Printer Commands Port Selection Font Selection Colour Management

COM3 Properties

Port Settings

Print to the following port:

Port:

COM2: ☐

COM5: ☐

COM8: ☐

COM9: ☐

COM6: ☐

COM7: ☐

COM3: ☒

Add Port...

Bits per second: 57600

Data bits: 8

Parity: None

Stop bits: 1

Flow control: Xon / Xoff

Restore Defaults

OK Cancel Apply

7. Embossing Layout Format

7.1. Embossing Format definition

The Embossing Format allows to define the following parameters for each line:

- Plate size
- X and Y line coordinate positions
- Font type
- Character spacing

Format can be up to 15 lines x 32 characters (columns).

Up to 3 Formats (from 0 to 2) can be store in the memory.

7.2. Motors Steps Ratio

Creating the format is good to know the step ratio of each motor:

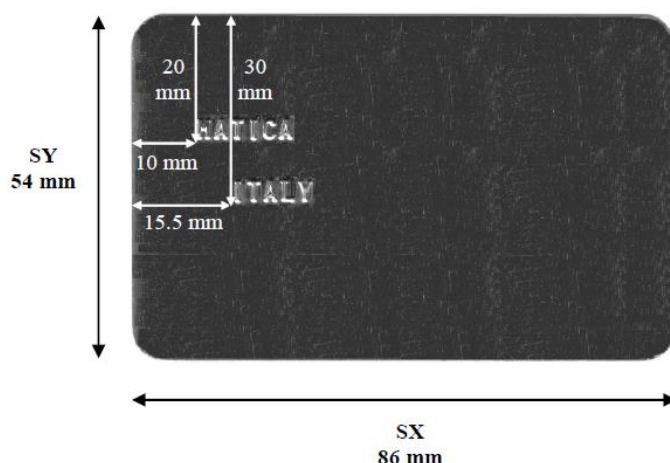
- Motor X step = 0,0907 mm (1/280 inch) → 1 mm = 11 steps
- Motor Y step = 0,125 mm → 1 mm = 8 steps
- Motor Drum = 2400 steps rotation → 1 char = 40 steps
- Motor Feeder step = 0.041 mm → 1 mm = 24 steps

7.3. Sample Embossing Format

This is a sample of embossing format:

```
<]F0 Y540 SX860 U0
Y200 X100 F0 CI10
Y300 X155 F0 CI10 >
```

7.4. How to measure the line coordinate



7.5. Format Header

I	Format Selector
F0	Format Number (mandatory) The format MUST BEGIN with Fn (n=0 to 3)
FN=name	Format Name (8 digits) <i>(non-mandatory)</i> The format name can be up to 8 digits. NO SPACES are allowed
SYxxx	Set the plate vertical size in 10 th mm if not specified will be used the default value in the settings (SY540)
SXxxx	Set the plate horizontal size 10 th mm if not specified will be used the default value in the settings (SX860)
Ux	Unit of Measure U0 = 10 th mm U1 = 100 th inch U2 = 1000 th inch U3 = Motor Step if not specified will be used the default value in the settings (U0 =mm/10)

7.6. Format Line Definition

N=name	Line Name (7 digits) The name can be up to 7 digits. NO SPACES are allowed The name is show using the off-line keyboard modality. For on-line use the name is not need.
Ynnn	Vertical Coordinate (mandatory) Set the embossing field position measured from the top edge of metal plate to the bottom edge of the character. The nnn value can be up to 4 digits and it is expressed in the Unit-of-Measure defined. ☞ NOTE: if the plate size (SYxxx) is not correct, the coordinate Ynnn will be wrong.
Xnnn	Horizontal Coordinate (mandatory) Set the embossing field position measured from the left edge of metal plate to the left edge of the character. The nnn value can be up to 4 digits and it is expressed in the Unit-of-Measure defined. ☞ note: "Y100 X100" or "X100 Y100" are both accepted.
Fn	Font Type Set the type of character to emboss: F0 = Font 0 F1 = Font 1 If not specified will be used the value of previous line. Default is F0 .
Clnn	Characters per Inch (cpi) Set the space between characters: CI10 = 10 characters per inch CI8 = 8 characters per inch If not specified it will be used the same value of previous line. Default is CI10 .
CSnn	Character Spacing (alternative of CI parameter in case of specific char spacing) Set the space between characters in number of steps: CS28 = 10 cpi standard for Simplex 2 CS35 = 8 cpi used for USA BLOCK

To facilitate the editing SPACE and CRLF are transparent and can be freely used.
 The following 2 formats are both accepted

```
<]F0 Y540SX860U0 Y200X100F0CI10 Y300X155F0CI10 >

<]F0 Y540 SX860 U0
Y200 X100 F0 CI10
Y300 X155 F0 CI10 >
```

8. Print Plates using Notepad

Create a new .TXT file and edit using Notepad.

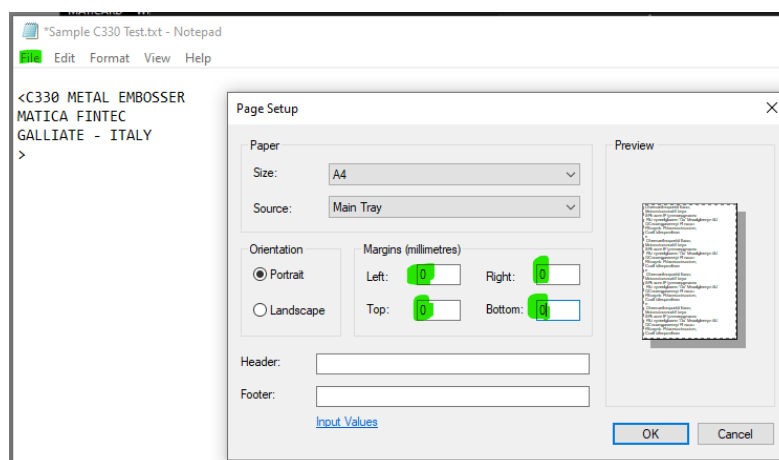
Do not use Notepad++ because don't allow to set the page margin and add the lines number.

```
<C330 METAL EMBOSSE
MATICA FINTEC ITALY
1234567890 >
```

Go to File\Page Setup and set ALL the margins at 0.

This to avoid having extra spaces on left side that will alter the X position.

This is a Notepad local setting and is need just one time because Notepad will save for next time.



Now Print on C330 printer.

Plate will start feeding and the above text will be embossed on the plate using the format 0 layout saved inside the C330 memory.

The embossing text the machine will use start from the symbol "<" and end with the symbol ">".

Data above the start-embossing "<" and after the end-embossing ">" are discharged.

We now set the embossing format layout editing:

```
<]F0 U0 SY540SX860
Y300X100 CI10
Y350X100 CI10
Y400X100 >
```

```
<C330 METAL EMBOSSE
MATICA FINTEC
GALLIATE - ITALY >
```

The first text starting with "<]F0" instruct the machine that we are sending an embossing format layout" (see chapter Embossing Format Layout for details).

You can send the format just one time and then all the plates text you need to produce.

9. ERROR LIST

When an error occurred, the display will show the messages listed below.

Apply the proper procedure to remove the error condition and then press CLEAR to continue (please carefully read the error comments).

E04 - X MOTOR ERROR (X-HOME)	During the movement to the hopper position something stops the carriage. Open the cover and check the cause.
E05 - Y MOTOR ERROR	Y movement error. Check the plate probably is wrong embossed, maybe the plate is bend and hit in the drum or the format set a Y coordinate too low.
E06 - X MOTOR ERROR (X-END)	During the movement to the unload position or during the embossing something stop the carriage. Check the plate probably is wrong embossed. If you heard a loud noise open the cover and remove the plate which is pre-feed.
E07 - DRUM MOTOR ERROR	Drum movement error. If you heard a loud noise open the cover and check manually if the drum is free to move; if not close the cover and press the SET key which will start the punch-recovery-procedure. Reopen the cover and verify that the drum is free to move, if OK press CLEAR to continue.
E08- PUNCH MOTOR ERROR	The punch cycle is lock. Press the SET key which will start the punch-recovery-procedure. When OK press CLEAR to continue.
E20 - FORMAT NUMBER ERROR (F0 to F9)	The format number must be from 0 to 9 for the embossing formats and from 10 to 19 for the Co-print formats.
E21- FORMAT NAME ERROR	The Format name is max 8 digit. A SPACE or CR must separate the format name to the next command F1 FN=TEST1 Y100X100 → OK F1 FN=TEST1Y100X100 → WRONG F1 FN=TEST 1 Y100X100 → WRONG
E22 - CARD DIMENSION SYNTAX ERROR	Wrong SX or SY command.
E23 - UNIT MEASUREMENT ERROR	Wrong Un command.
E24 - FIELD NAME ERROR	The Field name is max 7 digit. A SPACE or CR must separate the Field name to the next parameter N=LINE1 Y100X100 → OK N=LINE1Y100X100 → WRONG N=LINE 1 Y100X100 → WRONG
E25 - Y COORDINATE ERROR	Is OK: Y100 X100; Y50 X50; Y050 X050 Is WRONG: Y10 0 X100; Y 50 X50; Y 050 X050
E26 - X COORDINATE ERROR	Is OK: Y100 X100; Y50 X50; Y050 X050 Is WRONG: Y100 X10 0; Y50 X 50; Y050 X 050
E27 - TOO MANY FIELDS (max 50)	You exceed the maximum number of fields (50 max).
E28 - FONT ERROR	Use font 0 (F0) or font 1 (F1).
E29 - CHARACTER SPACE ERROR	Wrong Cinn or CSnn parameters.
E30 - VARIABLE FIELD SYNTAX ERROR	Check the syntax referring to the appropriate Manual chapter.
E31 - FIX FIELD SYNTAX ERROR	Check the syntax referring to the appropriate Manual chapter.
E32 - FORMAT WITHOUT FIELDS	The format needs at least 1 field to be used.
E33 - FIELD NOT COMPLETE	Check the field.
E34 - FIELD COMMAND ERROR	Command or Parameter wrong.
E35 - FORMAT MEMORY OVERFLOW	The format memory is over. Reediting the stored format and remove not needed Spaces in order to reduce the used memory.
E37 - FIELD-BUFFER OVERFLOW	You exceed the maximum number of characters.
E38 - ILLEGAL CHARACTER	A wrong character is received and cannot be emboss.
E44 - INPUT CONVERSION TABLE DATA LOST	Hardware error: the input conversion table is lost.
E48 - COVER OPEN ERROR	Hardware error: close the cover.

E64 - BUFFER OVFL.ERROR-DATA CORRUPTED	The buffer is overflow. The data are corrupted.
E70 - X MOTOR EMBOSS ERROR	X-axis card movement error; reject the card.
E71 - Y MOTOR EMBOSS ERROR	Y-axis card movement error; reject the card.
E72 - DRUM MOTOR EMBOSS ERROR	The card can even be picked or not by the embosser's clamp and the embossing sequence isn't completed correctly. If the clamp picks the card but doesn't start punching and the drum keeps on moving: a) Check drum motor home sensor; b) Drum motor home sensor is dirty: clean it with compressed air or lint free cloth; c) Drum motor home sensor isn't connected correctly on the board. If the card is picked, but it is embossed in a wrong way check: a) Belt tension; b) If pulleys are fixed on the shafts; c) If the motor is moving correctly or it stalls. If the card is picked by the clamp but the drum doesn't move check: a) Drum motor connection on the board.
E74 - CARD MISFEED	The clamp holds the card, but the embossing cycle doesn't start. Check the entry sensor: a) Entry sensor is dirty: clean it with compressed air or lint free cloth; b) Entry sensor isn't connected correctly on the board. The clamp moves straight to embossing area without a card or after having made a bad noise: a) Check for Y home sensor; b) Y home sensor is dirty: clean it with compressed air or lint free cloth; c) Y home sensor isn't connected correctly on the board; d) Check Y motor electrical connections; e) Check if the pulley is fixed on Y motor shaft; f) Check belt state.
E75 - XY CARD DIMENSION ERROR	Wrong plate dimensions.
E76 - XY COORDINATE ERROR	Wrong embossing coordinates.
E77 - LOAD POSITION ERROR	After correctly loaded, card fails to move to the destination position. There is a jam inside the module.
E78 - ILLEGAL CHARACTER	A wrong character cannot be embossed.
E80 - FEEDER HOME	Check the Feeder Home sensor.
E81 - FEEDER MOVEMENT	Check the Feeder Motor.
E82 - FEEDER ERROR	Feeder empty.